WRITTEN EXAM

LULEÅ UNIVERSITY OF TECHNOLOGY

Course Name: Mät- och reglerteknik / Measurement and Control	**Course Code:** R0005E
Number of examination hours: 4	**Date:** 2024-12-19

Allowed aids:

- non-programmed calculator
- an A4-sheet (both sides) of handwritten notes

On duty teacher (complete telephone number): Dina Shona Laila, +46 76 049 2718	**On duty teacher (complete telephone number):** Arman Sehatnia, +46 72 748 9994
On duty teacher (complete telephone number): Wolfgang Birk, +46 72 549 0909	**On duty teacher (complete telephone number):**

Grade scale:	**Number of questions and total score:**
3 – 12 points, 4 – 16 points, 5 – 20 points	4 questions, 24 points

Other information:

General instructions

Check that you have received all the exams/questions.
All new answers begin on a separate page. Do not write on the back page.
Use the provided semi-log paper to draw the Bode plot.
Print, write clearly, and do not use red pen.

After examination

Examination results will be announced within three weeks from the examination date and no later than two weeks before next re-examination.
The exam result can be seen at My LTU – Ladok for students.
Your corrected exam is scanned and visible on My LTU – Graded exams.

Information to the printer

Project number:	**Number of copies**	**Number of pages**
341980	21	3
Other information: Please attach the Appendices on each copy		

1. The dynamical motion of a translational mechanical system in Figure 1 is written as

$$\ddot{y}(t) + 5\dot{y}(t) + 2(y(t))^2 = F(t)$$

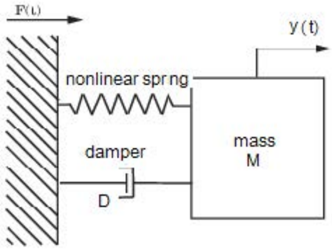


Figure 1: A translational mechanical system

(a) Find the linearized model of the system at one of its equilibria, where the force at that equilibrium is $F_0 = 0$. (2)

(b) Write the transfer function $G(s) = \frac{Y(s)}{F(s)}$ of the linearized model, and find its Laplace inverse. (2)

(c) Find the poles and zeros of the linearized model and determine the stability of the system. (1)

(d) Determine the steady state value of the unit step response of the liniearized model. (1)

Total for Question 1: 6 points

---

2. A model of an aeroplane pitch loop is given in Figure 2. The aircraft dynamics and the controller dynamics are respectively given by

$$G(s) = \frac{(s+10)}{(s^2+0.6s+9)} \quad \text{and} \quad D(s) = \frac{K}{(s+5)}.$$

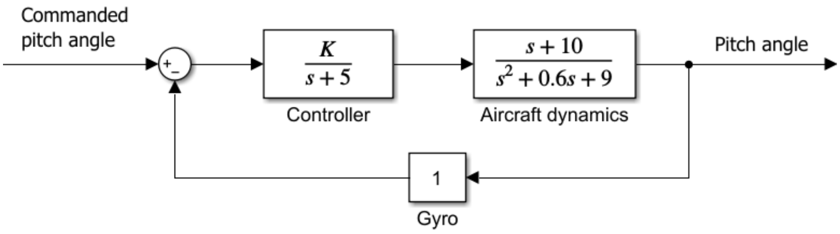


Figure 2: Block diagram of an aircraft pitch control loop.

(a) On the semilog graph paper(s) provided, sketch the Bode plot, both the magnitude and phase plots, of the open-loop transfer function $D(s)G(s)$, with $K = 1$. Include the sketch of the Bode plot of each term. (3)

(b) Explain why the Magnitude of a Bode plot is usually measured in dB. (0.5)

(c) Based on (a), determine the stability of the system. (1)

(d) Find the range of the value of the gain K for which the system is stable. (1.5)

Total for Question 2: 6 points

3. The open-loop transfer function of a system is given by

$$G(s) = \frac{K(s^2 - 8s + 15)}{s^2 + 2s + 8}$$

For $K = 1$, the Nyquist plot of $G(s)$ is shown in Figure 3.

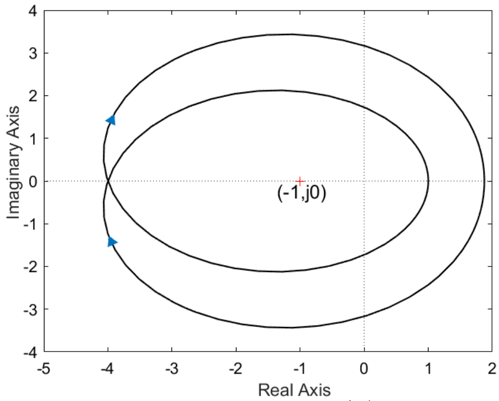


Figure 3: Nyquist plot for $G(s)$ with $K = 1$.

(a) Considering positive frequency, determine where the Nyquist plot starts (i.e. the value of $|G(j\omega)|$ and $\angle G(j\omega)$ when $\omega = 0$) and where the plot ends (i.e. the value of $|G(j\omega)|$ and $\angle G(j\omega)$ when $\omega = \infty$). (2)

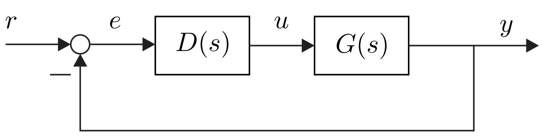
(b) Find the value of N, P and Z, and determine the stability of the closed-loop system using the Nyquist criterion. (2)

Total for Question 3: 4 points

4. The open-loop transfer function of a 1^{st} order plant is given by

$$G(s) = \frac{Y(s)}{U(s)} = \frac{3}{2s + 10}$$

A regulatory system must be developed for the process as follows



(a) A PI controller $(K_p + K_i/s)$ is added in series with the plant. Find the closed-loop transfer function of the system and write the closed-loop characteristic equation in terms of K_p and K_i. (2)

(b) Design the PI controller for the system so that the first pole of the closed-loop system is the same as the pole of the plant, and the second pole is twice of that first pole. (3)

(c) Determine the steady-state error of the response of the system, if the setpoint is a ramp signal with slope 1. (2)

(d) Explain what the expected effect of the use of the PI controller is to this system. (1)

Total for Question 4: 8 points

END

LULEÅ TEKNISKA UNIVERSITET

2.

KURSKOD / COURSE CODE	PROV / TEST CODE
R 0 0 0 5 E	0 0 0 5

KURSBENÄMNING / COURSE NAME

Mät- och reglerteknik

PROVBENÄMNING / TEST NAME

Tentamen

TENTAMENSDATUM / EXAMINATION DATE

2024-12-19

TENTAMENSORT/CITY (för distansstudenter / for off campus students only)

ÅÅÅÅMMDD-XXXX
YYYYMMDD-XXXX

[redacted]

PROGRAM | INSKR.ÅR/YEAR

[redacted]

ANTAL SIDOR / NO. OF PAGES: 08

Skanningsblad/Scanning Sheet

Behandlat uppgift nr (sätt x) / Mark the questions you answered with an X

Lärarens anteckningar / Teacher's notes

1	V	5.2	
2	V	1.7	
3	V	3.5	
4	V	7	
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			

Poängsumma Points	17.4	Betyg Grade	4

35065

LULEÅ TEKNISKA UNIVERSITET

Namn: [redacted] Personnummer: [redacted] Sidnr: 1

Uppgift nr: 1

Poäng: 4

Lärarens anteckning:

$\ddot{y}(t) + 5\dot{y}(t) + 2(y(t))^2 = F(t)$

a.) linearize; $F_0 = 0$

$y^2 \simeq y_0^2 + \frac{dy^2}{dy}\Big|_{y=y_0} \Delta y = y_0^2 + 2y_0 \Delta y$

$\Delta\ddot{y} + 5\Delta\dot{y} + 2y_0^2 + 4y_0\Delta y = F_0 + \Delta F$; we know $F_0 = 0 \Rightarrow$

2

$\Rightarrow \Delta\ddot{y} + 5\Delta\dot{y} + 2y_0^2 + 4y_0\Delta y = \Delta F$;

$\Delta = 0$ when $y = y_0 \Rightarrow \Delta y = y - y_0 \Rightarrow y_0 - y_0 = 0$

$2y_0^2 = 0 \Rightarrow y_0 = 0$; This gives the linearized system as $\Rightarrow$

$\Rightarrow \Delta\ddot{y} + 5\Delta\dot{y} = \Delta F$ ← answer

b.) we use Laplace: $\mathcal{L}\{\Delta F\} = \mathcal{L}\{\Delta\ddot{y} + 5\Delta\dot{y}\} \Rightarrow$

initial = 0

$\Rightarrow$ without beginning values $\Rightarrow F(s) = s^2Y(s) + 5sY(s) \Rightarrow$

2

$\Rightarrow F(s) = Y(s)(s^2 + 5s) \Rightarrow$

$\Rightarrow \frac{Y(s)}{F(s)} = \frac{1}{s(s+5)}$; $\frac{1}{s(s+5)} = \frac{A}{s} + \frac{B}{s+5} \Rightarrow$

$\Rightarrow 1 = A(s+5) + B(s) \Rightarrow 1 = As + 5A + Bs$

$s^1: 0 = A + B \Rightarrow A = -B$

$s^0: 1 = 5A \Rightarrow A = \frac{1}{5} \Rightarrow B = -\frac{1}{5}$

$\frac{1}{s(s+5)} = \frac{1}{5}\cdot\frac{1}{s} - \frac{1}{5}\cdot\frac{1}{s+5}$; $\mathcal{L}^{-1}\{\frac{1}{5}\cdot\frac{1}{s} - \frac{1}{5}\cdot\frac{1}{s+5}\} = \frac{1}{5}\cdot 1 - \frac{1}{5}\cdot e^{-5t}$

so: Laplace inverse $= \frac{y(t)}{F(t)} = \frac{1}{5}(1 - e^{-5t})$

Namn: [redacted] Personnummer: Sidnr: 2

Uppgift nr: 1

Poäng: 1.2

Lärarens anteckning: 1

c.) zeroes: there are no zeroes ✓

poles; $s(s+5)=0 \Rightarrow \begin{matrix} s=-5 \\ s=0 \end{matrix}$ ✓

one pole on left hand plane, and one on $s=0$ wich makes the system marginally stable ✓

d.) steady state value: $\lim_{t\to\infty} y(t)$

$$\lim_{t\to\infty} \frac{1}{5}(1-e^{-5t}) = \frac{1}{5}(1-0) = \frac{1}{5}$$

need to include unit step input

0.2

$$\frac{Y}{F} = \frac{1}{s(s+5)} \qquad F = \frac{1}{s}$$

$$Y = \frac{1}{s} \cdot \frac{1}{s(s+5)} = \frac{1}{s^2(s+5)}$$

SS value, use FVT

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} \cancel{s} \frac{1}{s^{\cancel{2}}(s+5)} = \frac{1}{0} = \infty$$

Luleå tekniska universitet, Grafisk produktion

Namn:	Personnummer:	Sidnr:
[redacted]		3

Uppgift nr: 2

Poäng: 0.2

Lärarens anteckning: 0.2

$$G(s) = \frac{s+10}{s^2+0{,}6s+9} \quad ; D(s) = \frac{K}{s+5}$$

a.)

open loop with $K=1$; $T(s) = G(s) \cdot D(s) =$

$$= \frac{s+10}{(s^2+0{,}6s+9)} \cdot \frac{1}{(s+5)} = \frac{s+10}{s^3+0{,}6s^2+9s+5s^2+3s+45} =$$

why do this ??

$$= \frac{s+10}{s^3+5{,}6s^2+12s+45} = T(s)$$

$$T(jw) = \frac{jw+10}{-jw^3-5{,}6w^2+12jw+45}$$

$$Re\{T(jw)\} = \frac{10}{-5{,}6w^2+45}$$

$$Im\{T(jw)\} = \frac{w}{-w^3-5{,}6w^2+12w}$$

I dont know how to plot in this manner but i can still analyze the system:

$$\lim_{w \to \infty} \frac{jw+10}{-jw^3-5{,}6w^2+12jw+45} = 0$$

?

$$\lim_{w \to 0} \frac{jw+10}{-jw^3-5{,}6w^2+12jw+45} = \frac{10}{45}$$

$$T(0) = \frac{10+0j}{45} = \frac{10}{45}+0j \Rightarrow \angle T(jw) = 0^\circ$$

$$T(\infty) = \frac{jw}{-jw^3-5{,}6w^2+12jw} = 0 + \frac{1}{12}j \Rightarrow \angle T(jw) = 90^\circ$$

Namn: [redacted] Personnummer: [redacted] Sidnr: 4

Uppgift nr: 2

Poäng: 1

Lärarens anteckning:

b.) we measure it in dB because its an exponential curve. this gives a good and readable graph rather than if we didnt use dB (dB is log, wich is exponential) No! this is for the use of log frequency! ∅

c.) 0 dB is never crossed => PM = ∞
the phase shift approaches -180° as w tends to be ∞, therfore the GM is also infinite ✓ OK, but not accurate 1
when PM>0 & GM>0 the system is stable

d.) T(S) with K => $T(s) = \frac{G(s) \cdot D(s)}{1 + G(s) \cdot D(s) \cdot Gyro} =$
(closed loop)

$$= \frac{\frac{(s+10)}{(s^2+0,6s+9)} \cdot \frac{(K)}{(s+5)}}{1 + \frac{s+10}{s^2+0,6s+9} \cdot \frac{K}{(s+5)}} = \frac{\frac{K(s+10)}{(s+5)(s^2+0,6s+9)}}{1 + \frac{K(s+10)}{(s+5)(s^2+0,6s+9)}} =$$

$$= \frac{\frac{K(s+10)}{\cancel{(s+5)(s^2+0,6s+9)}}}{\frac{(s+5)(s^2+0,6s+9)+K(s+10)}{\cancel{(s+5)(s^2+0,6s+9)}}} = \frac{K(s+10)}{(s+5)(s^2+0,6s+9)+K(s+10)} =$$

$$= \frac{K(s+10)}{s^3+0,6s^2+9s+5s^2+3s+45+Ks+10K} =$$

$$= \frac{K(s+10)}{s^3+5,6s^2+9s+Ks+45+10K} = \frac{K(s+10)}{s^3+5,6s^2+(9+K)s+(45+10K)}$$

Next Page

Namn: [redacted] Sidnr: 5

Uppgift nr: 2

Poäng: 0.5

Lärarens anteckning:

d. continues)

We have: $s^3+5{,}6s^2+(9+K)s+(45+10K)=0$

Routh Array:

s^3	1	$9+K$
s^2	$5{,}6$	$45+10K$
s^1	$\frac{5{,}4-4{,}4K}{5{,}6}$	0
s^0	$45+10K$	0

$$\frac{5{,}6\cdot(9+K)-(45+10K)\cdot 1}{5{,}6} = \frac{50{,}4+5{,}6K-45-10K}{5{,}6} = \frac{5{,}4-4{,}4K}{5{,}6}$$

$$\frac{\frac{5{,}4-4{,}4K}{5{,}6}\cdot(45+10K)-5{,}6\cdot 0}{\frac{5{,}4-4{,}4K}{5{,}6}} = 45+10K$$

For the system to be stable we want no sign change in first collum.

0.5

$$\frac{5{,}4-4{,}4K}{5{,}6}\geq 0 \Rightarrow 5{,}4-4{,}4K\geq 0 \Rightarrow -4{,}4K\geq -5{,}4 \Rightarrow$$
$$\Rightarrow K\leq 1{,}227$$

$$45+10K\geq 0 \Rightarrow K\geq -4{,}5$$

System is stable when $1{,}227>K>-4{,}5$

Answer: range of values for K is

What about te effect of zero!?

Use Bode plot Gain instead!

Namn: [redacted] Personnummer: Sidnr: 6

Uppgift nr: 3

Poäng: 3.5

Lärarens anteckning:

$$G(s) = \frac{K(s^2-8s+15)}{s^2+2s+8} \quad ; K=1 \text{ in this case}$$

a.) $G(jw) = \dfrac{-w^2-8jw+15}{-w^2+2jw+8} \Rightarrow \dfrac{(-w^2-8jw+15)}{(-w^2+2jw+8)}\dfrac{(-w^2-2jw+8)}{(-w^2-2jw+8)}$

$$\Rightarrow G(jw) = \frac{w^4+2jw^3-8w^2+8jw^3-16w^2-64jw-15w^2-30jw+120}{w^4+2jw^3-8w^2-2jw^3+4w^2+16jw-8w^2-16jw+64}$$

$$\Rightarrow G(jw) = \frac{w^4+10jw^3-39w^2-94jw+120}{w^4-12w^2+64}$$

$$Re\{G(jw)\} = \frac{w^4-39w^2+120}{w^4-12w^2+64};$$

$$Im\{G(jw)\} = \frac{10w^3-94w}{w^4-12w^2+64};$$

I think this should be 180°...

why? 2

$G(0) = \dfrac{120+0j}{64} = 1{,}875+0j$; $\angle G(jw) = 0°$

$G(\infty) = \lim_{w\to\infty} \dfrac{w^4}{w^4} + j\dfrac{10w^3}{w^4} = 1+0j$ $\angle G(jw) = 0°$ 360

1.5

b.) Find: N, P, Z : poles of $G(s) = s^2+2s+8 \Rightarrow$

$\Rightarrow$ PQ: $s = -\frac{2}{2} \pm \sqrt{(\frac{2}{2})^2-8} \Rightarrow s = -1 \pm \sqrt{1-8}$

Both LHP, $P=0$

1 CW encirclement of $(-1,0j) \Rightarrow N = \cancel{1}$ 2 times

$Z = N+P \Rightarrow Z = 1+0$;

There is one pole in RHP, therefore the system is unstable

Luleå tekniska universitet

Namn: [redacted] Personnummer: [redacted]

Sidnr: 7

Uppgift nr: 4

Poäng: 5

Lärarens anteckning:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{3}{2s+10} \text{ ; PI added} \Rightarrow D(s) = K_p + \frac{K_i}{s}$$

a.) closed loop: $T(s) = \frac{G(s)\cdot D(s)}{1+G(s)\cdot D(s)} = \frac{\left(K_p+\frac{K_i}{s}\right)\left(\frac{3}{2s+10}\right)}{1+\left(K_p+\frac{K_i}{s}\right)\left(\frac{3}{2s+10}\right)} =$

$$= \frac{\left(\frac{K_p\cdot s+K_i}{s}\right)\left(\frac{3}{2s+10}\right)}{1+\left(\frac{K_p\cdot s+K_i}{s}\right)\left(\frac{3}{2s+10}\right)} = \frac{\frac{3(K_p\cdot s+K_i)}{s(2s+10)}}{1+\frac{3(K_p\cdot s+K_i)}{s(2s+10)}} = \frac{\frac{3(K_p\cdot s+K_i)}{s(2s+10)}}{\frac{s(2s+10)+3(K_p s+K_i)}{s(2s+10)}} =$$

$$= \frac{3(K_p s+K_i)}{s(2s+10)+3(K_p s+K_i)} = \frac{3(K_p\cdot s+K_i)}{2s^2+10s+3K_p\cdot s+3K_i} =$$

2

$$= \frac{3(K_p\cdot s+K_i)}{2s^2+(10+3K_p)\cdot s+3K_i} \text{ . Characteristic equation} \Rightarrow$$

$$\Rightarrow 2s^2+(10+3K_p)s+3K_i = 0$$

b.) plant: $G(s) = \frac{3}{2s+10} \Rightarrow$ Pole: $2s+10=0 \Rightarrow s_1=-5$

second pole: $2\cdot -5 = -10$

$s_2 = -10$

We get: $(2s+10)(s+10) = 2s^2+20s+10s+100 =$

$= 2s^2+30s+100$

Set this equal to char ekr from a.) and solve for K_p & K_i

3

$2s^2+(10+3K_p)s+3K_i = 2s^2+30s+100$

$10+3K_p = 30 \Rightarrow 3K_p = 20 \Rightarrow K_p = \frac{20}{3}$

$3K_i = 100 \Rightarrow K_i = \frac{100}{3}$

PI controller: $D(s) = \frac{1}{3}\left(20+\frac{100}{s}\right)$

Luleå tekniska universitet, Grafisk produktion

LULEÅ TEKNISKA UNIVERSITET

Namn: [redacted] Personnummer: [redacted] Sidnr: 8

Uppgift nr: 4

Poäng: 2

Lärarens anteckning:

c.) determine steady-state error: $\lim_{t \to 0} y(t)$

$$T(s) = \frac{3(K_p \cdot s + K_i)}{2s^2 + (10+3K_p)s + 3K_i} \Rightarrow K_p = \frac{20}{3}, \; K_i = \frac{100}{3} \Rightarrow$$

$$\Rightarrow T(s) = \frac{3\left(\frac{20}{3} \cdot s + \frac{100}{3}\right)}{2s^2 + \left(10 + 3 \cdot \frac{20}{3}\right)s + 3 \cdot \frac{100}{3}} = \frac{20s+100}{2s^2+30s+100}$$

We know from Laplace: $\lim_{t \to 0} y(t) = \lim_{s \to \infty} s \cdot Y(s)$

$e_{ss} = \lim_{s \to \cancel{\infty}} s \cdot E(s)$; $E(s) = \cancel{(1 - Y(s)) \cdot R(s)}$ $T(s)$

$E(s) = R(s) - Y(s)$
$R(s) - T(s)R(s) = R(s)(1 - T(s))$

$R(s) = \frac{1}{s^2}$ ← ramp with slope 1

$$e_{ss} = \lim_{s \to \cancel{\infty}} \cancel{s} \cdot \left(1 - \frac{20s+100}{2s^2+30s+100}\right) \cdot \frac{1}{s^{\cancel{2}}} =$$

1

$$= \lim_{\cancel{s \to \infty}} \left(\frac{1}{s} - \frac{1}{s}\left(\frac{20s+100}{2s^2+30s+100}\right)\right) = \lim_{\cancel{s \to \infty}} \left(\frac{1}{s} - \frac{20s+100}{2s^3+30s^2+100s}\right) =$$

$s \to 0$ $s \to 0$

$= 0 - 0 = 0$

correct derivation, but wrong limit!

d.) The expected result of the PI controller is a faster response time, better tracking of the input, and better load disturbance rejection

1